TASK S3.01

Table Manipulation

Level 1

- Exercise 1

La teva tasca és dissenyar i crear una taula anomenada "credit_card" que emmagatzemi detalls crucials sobre les targetes de crèdit. La nova taula ha de ser capaç d'identificar de manera única cada targeta i establir una relació adequada amb les altres dues taules ("transaction" i "company").

Challenges and solution overview

a.- Table creation

The credit_card table is created with id as the primary key to ensure unique identification of each credit card.

b.- Data types for sensitive fields

The pin and cvv fields are defined as CHAR instead of numeric types. These fields are not used for calculations, and using CHAR prevents the loss of leading zeros.

The cvv field is defined with four characters to support AMEX cards, which may use four-digit CVV values.

```sql
CREATE TABLE IF NOT EXISTS credit_card (
    id VARCHAR(15) PRIMARY KEY,
    iban VARCHAR(50),
    pan VARCHAR(30),
    pin CHAR(4),
    cvv CHAR(4),  -- reservo 4 bytes para futuro uso con AMEX
    expiring_date VARCHAR(10)
);
```

100% 3:23

ction Output

	Time	Action	Response	Duration / Fe
1	18:18:08	CREATE TABLE I...	0 row(s) affected	0.040 sec

After creating the table 'credit_card' we import the data from the file “datos_introducir_sprint3_credit”.

c.- Data import and ‘date’ normalization

From my point of view in these exercises we must maintain and follow the good practice of the analyst whenever possible. For this reason in this case even if the exercise does not request it and also in Level 3 we will have to revert, here I have decided to standardize the field "expiring_date".

The expiration date provided in the dataset is not in standard SQL date format. Therefore, the expiring_date column is initially defined as VARCHAR(10) and later converted to a DATE type using STR_TO_DATE.

The original format is (MM/DD/YY), which was validated by detecting the leap year 2028 (02/29/28).

To ensure data integrity, an intermediate column (expiring_date_temp) will be used during the conversion process and then renamed as the final expiring_date column.

First I present a screenshot of the entire transformation process, and in the next page I show the process step by step with its executions.

```sql
-- Fix problem with 'expiring_date' column
-- New Column creation
-- Copy values to new column formating to DATE. alter
-- I use WHERE with 'expiring_date' but to avoid
-- MyWorkBench security protection I deactivated and reactivated Safe Updates

ALTER TABLE credit_card
ADD expiring_date_temp DATE
;

SET SQL_SAFE_UPDATES = 0;

UPDATE credit_card
SET expiring_date_temp = STR_TO_DATE(expiring_date, '%m/%d/%Y')
WHERE expiring_date IS NOT NULL;

SET SQL_SAFE_UPDATES = 1;

-- Data checking in order to erase original column and rename the new one
SELECT expiring_date, expiring_date_temp
FROM credit_card
LIMIT 10;

-- Drop the column and rename
ALTER TABLE credit_card
DROP COLUMN expiring_date
;
ALTER TABLE credit_card
RENAME COLUMN expiring_date_temp TO expiring_date
;
```

As I have commented before, this whole part is an Off Topic that does not ask for the statement, but as we are here to learn I have decided to proceed with the transformation.

Step by Step

New temporary column creation on crecit_card table

```
-- 2 - Fix problem with 'expiring_date' column
-- New Column creation
-- Copy values to new column formating to DATE. alter
-- I use WHERE with 'expiring_date' but to avoid
-- MyWorkBench security protection I deactivated and reactivated Safe Updates

ALTER TABLE credit_card
ADD expiring_date_temp DATE
;
```

100% 24:32

Action Output

	Time	Action	Response	Duration
1	22:07:00	ALTER TABLE transaction AD...	100000 row(s) affected Records: 100000 Duplicates:	0.606 se

Use of the Function STR_TO_DATE to convert the non standardized date to an standardized one!

To proceed has been necessary to temporarily deactivate the MySQL protection for massive changes.

```
SET SQL_SAFE_UPDATES = 0;

UPDATE credit_card
SET expiring_date_temp = STR_TO_DATE(expiring_date, '%m/%d/%Y')
WHERE expiring_date IS NOT NULL;

SET SQL_SAFE_UPDATES = 1;
```

100% 26:42

Action Output

	Time	Action	Response
1	18:55:06	SET SQL_SAFE_UPDATES = 0	0 row(s) affected
2	18:55:13	UPDATE credit_card SET expiring_date_temp = S...	5000 row(s) affected Rows matched: 5000 Changed: 5000 W
3	18:55:23	SET SQL_SAFE_UPDATES = 1	0 row(s) affected

Check that the procedure has worked properly

```sql
-- Data checking in order to erase original column and rename the new one
SELECT expiring_date, expiring_date_temp
FROM credit_card
LIMIT 10;
```

100% 74:44

Result Grid Filter Rows: Search Export:

expiring_date	expiring_date_temp
09/27/25	2025-09-27
12/28/28	2028-12-28
11/26/26	2026-11-26
07/27/27	2027-07-27
04/25/26	2026-04-25
11/27/26	2026-11-27
12/27/29	2029-12-27
02/28/26	2026-02-28
11/25/28	2028-11-25
02/28/25	2025-02-28

credit_card 1

ction Output

	Time	Action	Response	Duration / Fetch T
4	18:58:58	SELECT expiring_date, expiring_date_temp FRO...	10 row(s) returned	0.011 sec / 0.0006

Final process, we remove the initial column and rename the transformed column.

```sql
-- Drop the column and rename
ALTER TABLE credit_card
DROP COLUMN expiring_date
;
ALTER TABLE credit_card
RENAME COLUMN expiring_date_temp TO expiring_date
;
```

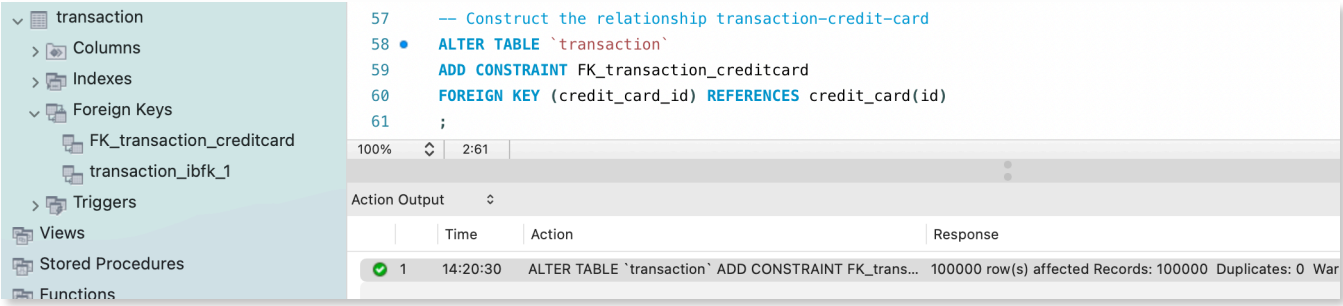
100% 2:55

Action Output

	Time	Action	Response	Duration / Fetch Time
1	22:03:28	ALTER TABLE credit_card DROP COL...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.029 sec
2	22:03:43	ALTER TABLE credit_card RENAME C...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.0063 sec

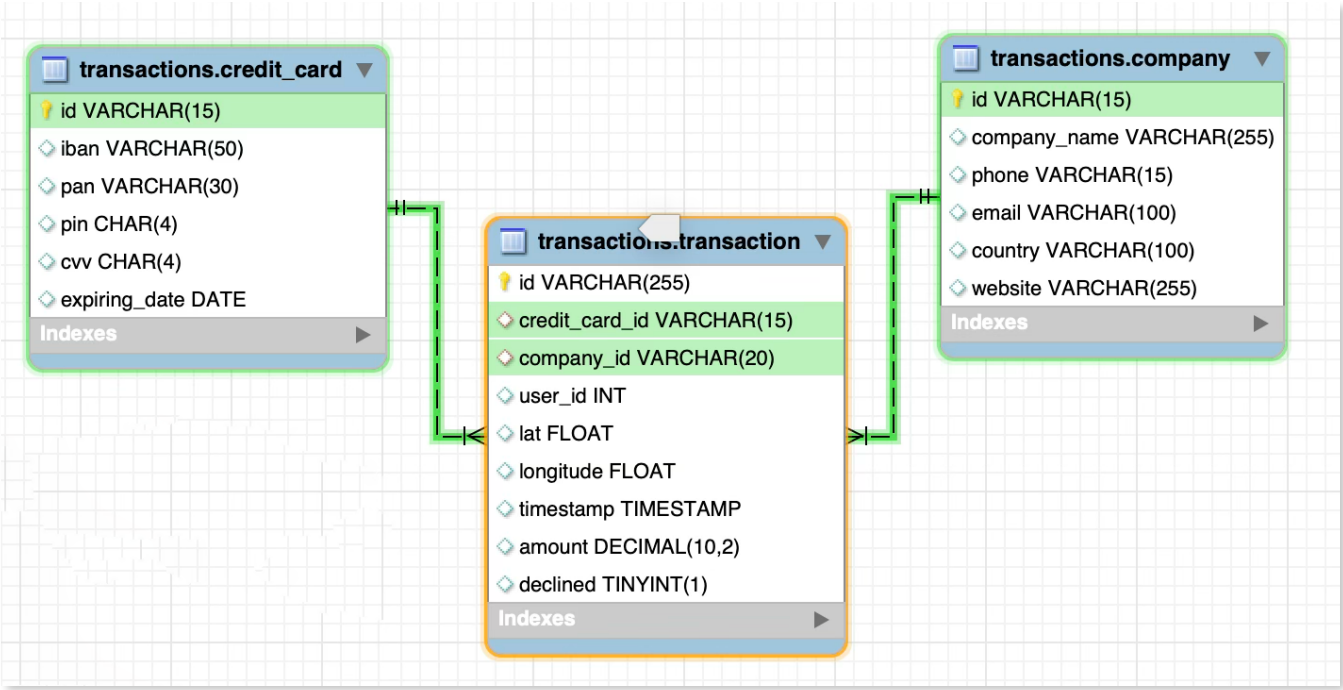
Després de crear la taula serà necessari que ingressis la informació del document denominat "dades_introduir_credit". Recorda mostrar el diagrama i realitzar una breu descripció d'aquest.

d.- Table relationships



We have the 'transaction' table as the fact table, while the other two tables act as dimension tables. At this stage, the database follows a **star schema dimensional structure**.

We can clearly identify that 'transaction has a Primary Key that is the 'id' of each transaction and also 3 Foreign Key that relate it or will relate it to a carnality of "N" many in 'transaction' to one in the dimension tables.



- Exercise 2

El departament de Recursos Humans ha identificat un error en el número de compte associat a la targeta de crèdit amb ID CcU-2938. La informació que ha de mostrar-se per a aquest registre és: TR32345631221357681769999 9. Recorda mostrar que el canvi es va realitzar.

CcS-9580	XX781258889851950806677358	5541182364498931	9273	737	2029-03-27
CcS-9581	XX9156705164053881243981 47	2624305470167630	4336	926	2025-06-29
CcU-2938	TR30195031221357681763866 1	5424465566813633	3257	984	2022-10-30
CcU-2945	DO2685476374853747521656868 9	5142423821948828	9080	887	2023-08-24
CcU-2952	BG45IVQL52710525608255	4556 453 55 5287	4598	438	2021-06-29
CcU-2959	CR7242477244335841535	372461377349375	3583	667	2023-02-24

```
70 •  UPDATE credit_card
71    SET iban = 'TR32345631221357681769999 9'
72    WHERE id = 'CcU-2938'
73    ;
74
```

100% 2:73

Action Output

	Time	Action	Response
1	10:14:00	UPDATE credit_card SET iban = 'TR32345631221357681769999 9' WHERE id = 'CcU-2938'	1 row(s) affected Rov

CcS-9580	XX781258889851950806677358	5541182364498931	9273	737	2029-03-27
CcS-9581	XX9156705164053881243981 47	2624305470167630	4336	926	2025-06-29
CcU-2938	TR32345631221357681769999 9	5424465566813633	3257	984	2022-10-30
CcU-2945	DO2685476374853747521656868 9	5142423821948828	9080	887	2023-08-24
CcU-2952	BG45IVQL52710525608255	4556 453 55 5287	4598	438	2021-06-29

- Exercise 3

En la taula "transaction" ingressa una nova transacció amb la següent informació:

Id	108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id	CcU-9999
company_id	b-9999
user_id	9999
lat	829.999
longitude	-117.999
amount	111.11
declined	0

I want to run the query at once, inserting the values on the PK of the dimensional tables if they are not there, to check this we use the clause “ON DUPLICATE KEY UPDATED”. I will use START TRANSACTION - COMMIT that for me it looks a very interesting way

```
START TRANSACTION;

INSERT INTO credit_card (id)
VALUES ('CcU-9999')
ON DUPLICATE KEY UPDATE id = id;

INSERT INTO company (id)
VALUES ('b-9999')
ON DUPLICATE KEY UPDATE id = id;

INSERT INTO `transaction` (
    id, credit_card_id, company_id, user_id, lat, longitude, amount, declined)
VALUES (
    '108B1D1D-5B23-A76C-55EF-C568E49A99DD', 'CcU-9999', 'b-9999', 9999, 829.999, -117.999, 111.11, 0);

COMMIT;
```

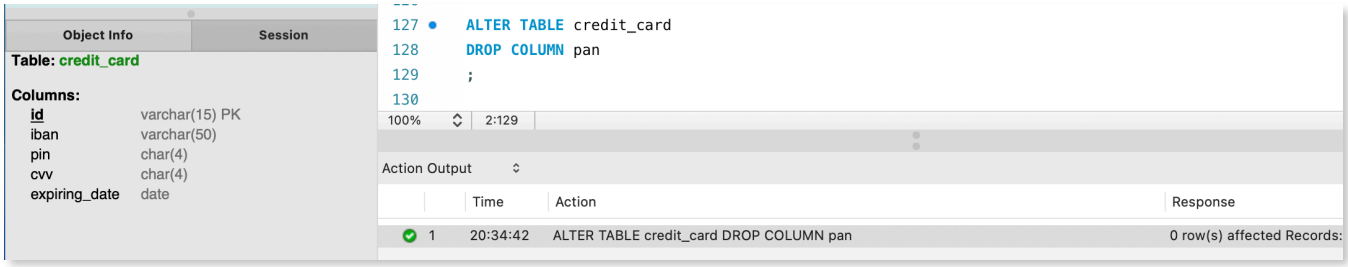
100% 4:88

Action Output

	Time	Action	Response
1	14:26:39	START TRANSACTION	0 row(s) affected
2	14:26:39	INSERT INTO credit_card (id) VALUES ('CcU-9999') ON...	1 row(s) affected
3	14:26:39	INSERT INTO company (id) VALUES ('b-9999') ON DUPL...	1 row(s) affected
4	14:26:39	INSERT INTO `transaction` (id, credit_card_id, compan...	1 row(s) affected
5	14:26:39	COMMIT	0 row(s) affected

- Exercise 4

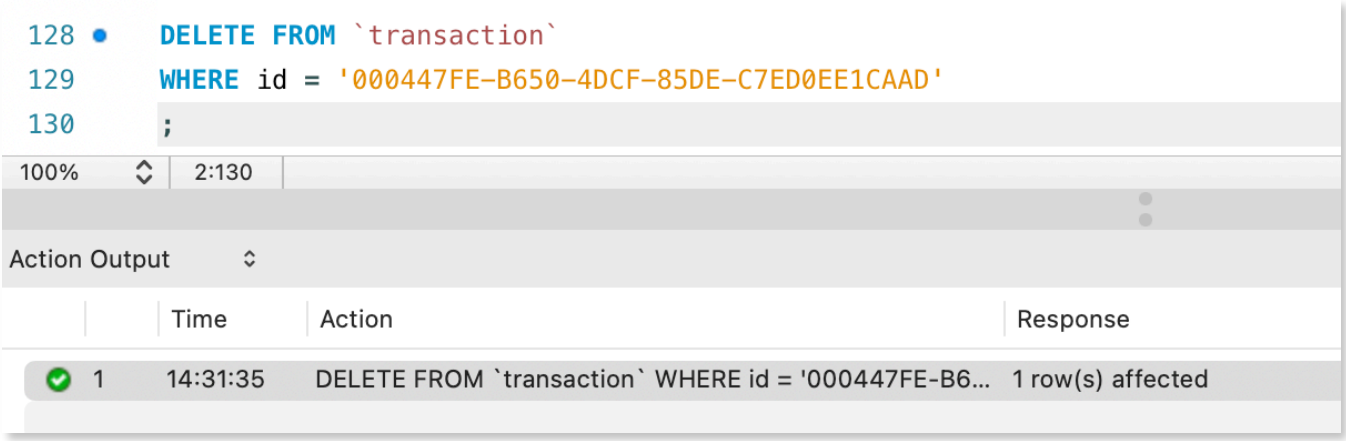
Des de recursos humans et sol·liciten eliminar la columna "pan" de la taula credit_card. Recorda mostrar el canvi realitzat.



Level 2

- Exercise 1

Elimina de la taula transaction el registre amb ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD de la base de dades.



- Exercise 2

La secció de màrqueting desitja tenir accés a informació específica per a realitzar anàlisi i estratègies efectives. S'ha sol·licitat crear una vista que proporcioni detalls clau sobre les companyies i les seves transaccions. Serà necessària que creïs una vista anomenada VistaMarketing que contingui la següent informació: Nom de la companyia. Telèfon de contacte. País de residència. Mitjana de compra realitzat per cada companyia. Presenta la vista creada, ordenant les dades de major a menor mitjana de compra.

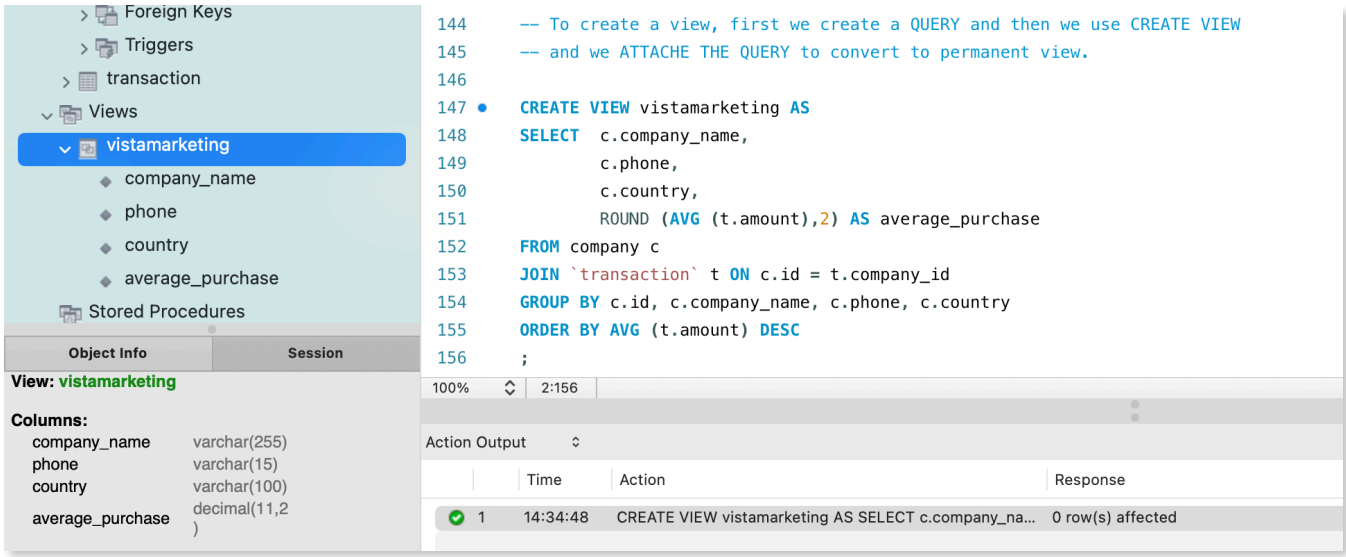
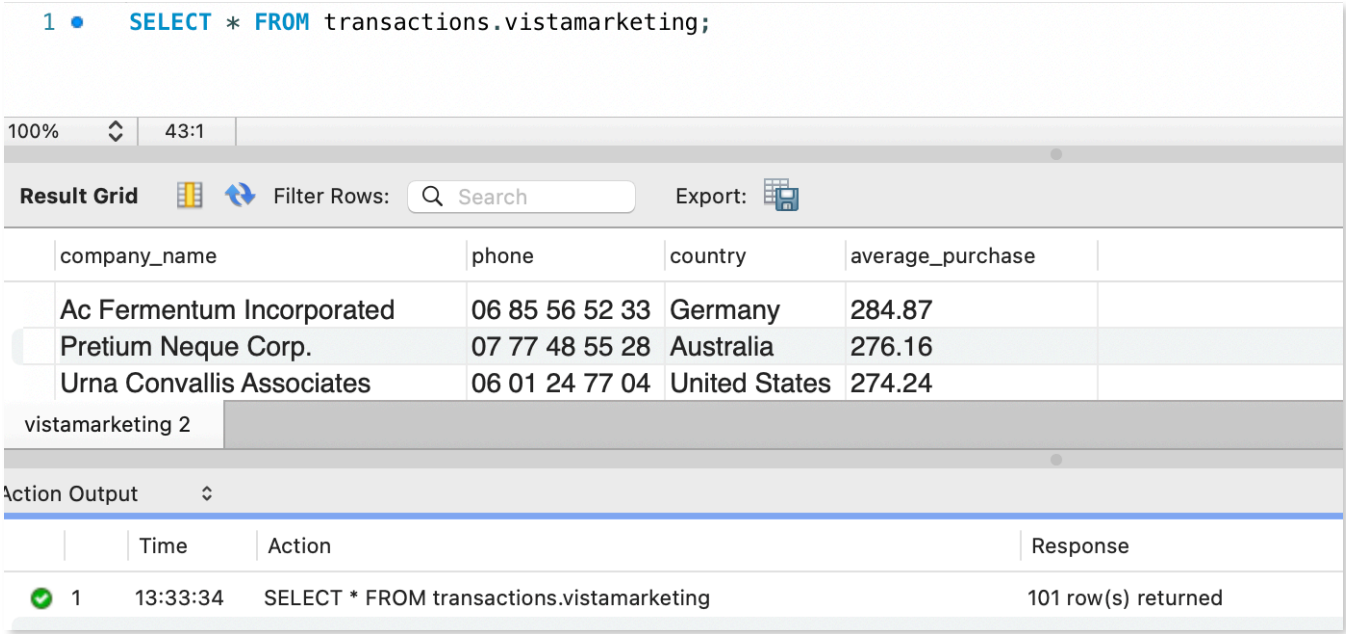


Tabla resultante de ejecutar la view ‘vistamarketing’



- Exercise 3

Filtra la vista VistaMarketing per a mostrar només les companyies que tenen el seu país de

```
174 •  SELECT *
175    FROM vistamarketing
176    WHERE country = 'Germany'
177    ;
```

100% 2:177

Result Grid Filter Rows: Search Export:

company_name	phone	country	average_purchase
Ac Fermentum Incorporated	06 85 56 52 33	Germany	284.87
Nunc Interdum Incorporated	05 18 15 48 13	Germany	259.32
Convallis In Incorporated	06 66 57 29 50	Germany	257.75
Ac Industries	09 34 65 40 60	Germany	255.15
Rutrum Non Inc.	02 66 31 61 09	Germany	255.14
Auctor Mauris Corp.	05 62 87 14 41	Germany	254.77
Augue Foundation	06 88 43 15 63	Germany	253.51
Aliquam PC	01 45 73 52 16	Germany	253.14

vistamarketing 6

Action Output

	Time	Action	Response
1	21:37:43	SELECT * FROM vistamarketing WHERE country = 'Germany'	8 row(s) returned

Level 3

- Exercise 1

La setmana vinent tindràs una nova reunió amb els gerents de màrqueting. Un company del teu equip va realitzar modificacions en la base de dades, però no recorda com les va realitzar. Et demana que l'ajudis a deixar els comandos executats per a obtenir el següent diagrama:

Recordatori:

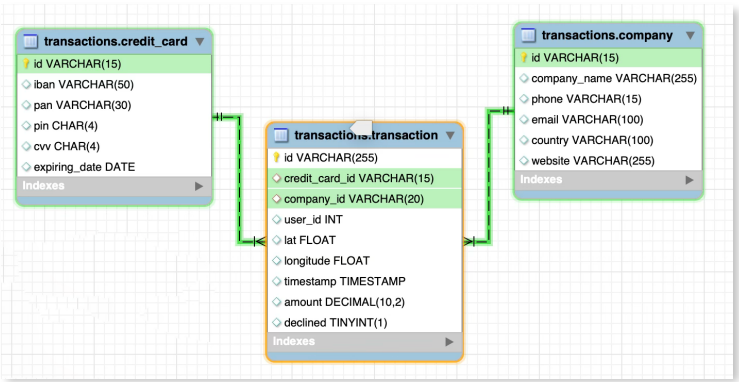
En aquesta activitat, és necessari que descriguis el "pas a pas" de les tasques realitzades. És important realitzar descripcions senzilles, simples i fàcils de comprendre. Per a realitzar aquesta activitat hauràs de treballar amb els arxius denominats "estructura_dades_user" i "dades_introduir_user"

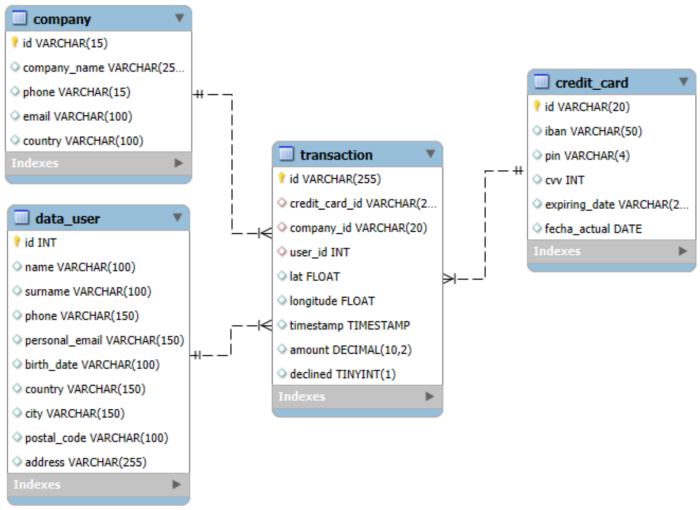
Recorda continuar treballant sobre el model i les taules amb les quals ja has treballat fins ara.

Objective:

Execute commands to obtain diagram 2 from diagram 1. Describe the transformation process in detail.

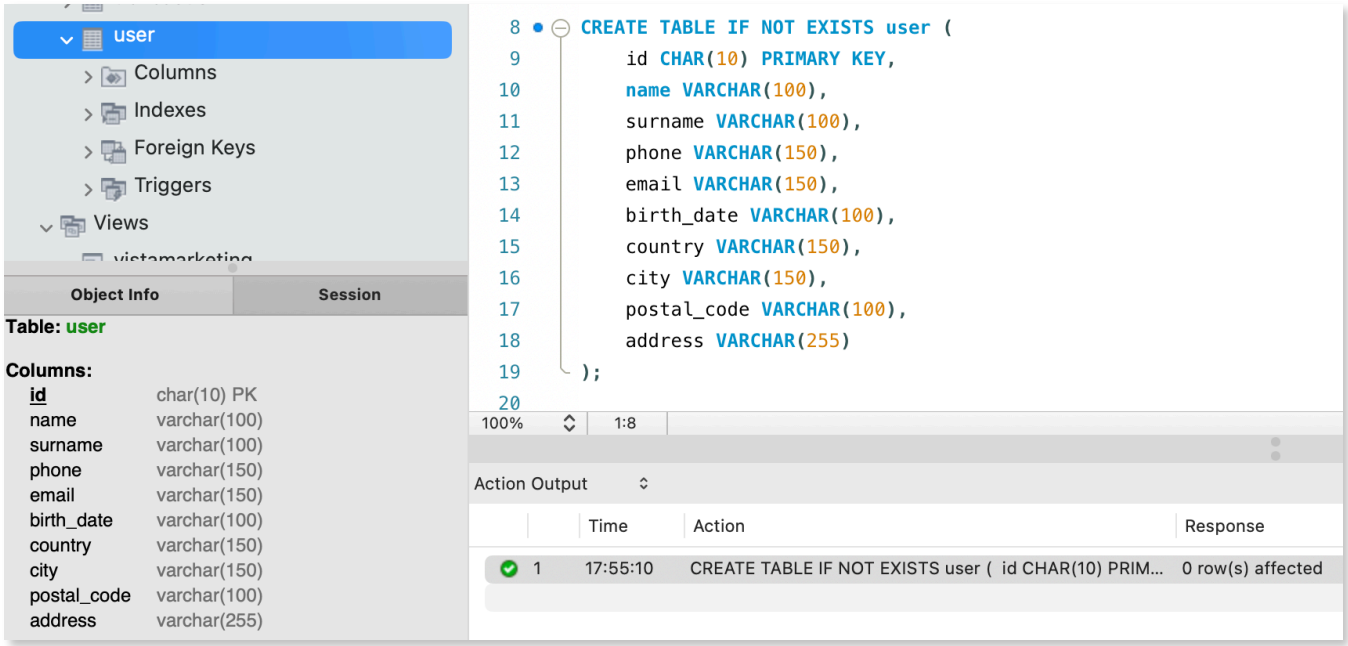
DIAGRAM 1 → ***DIAGRAM 2***



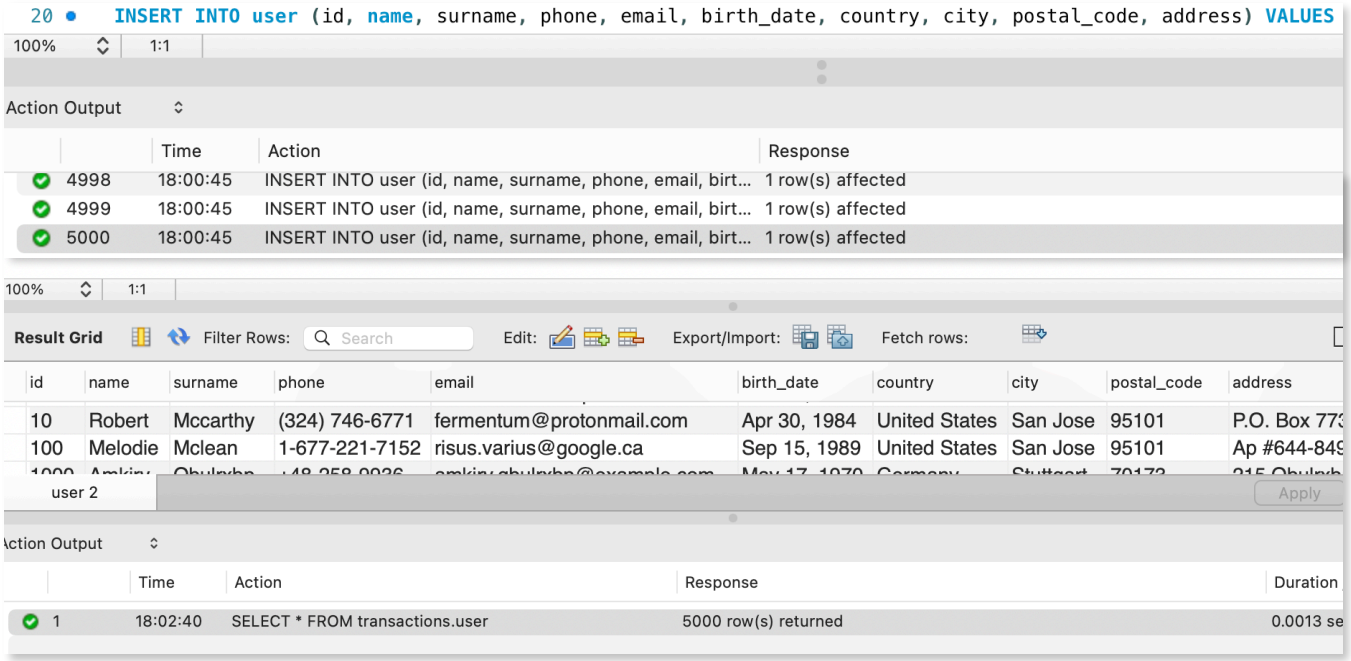


Step 1:

- Open in MySQL Workbench or the appropriate IDE the file: 'estructura datos user.sql'.
- Execute the script to create table 'user'.



- Open in your IDE the file with the data for 'user' table: 'datos introducir sprint3 user.sql'.
- Execute the script to insert the data in the table 'user'.



Step 2: Rename table 'user' —> 'data_user'

```
22 •  ALTER TABLE `user`
23    RENAME TO data_user
24    ;
```

100% 1:18

Action Output

	Time	Action	Response
1	18:28:24	ALTER TABLE `user` RENAME TO data_user	0 row(s) affected

Step 3: Change column type 'id' from table 'data_user' to INT

```
29 •  ALTER TABLE data_user
30    MODIFY COLUMN id INT
31    ;
```

100% 2:31

Action Output

	Time	Action	Response
1	18:40:58	ALTER TABLE data_user MODIFY COLUMN id INT	5000 row(s) affected Records: 5000 Duplicates: 0 Warnings: 0

Step 4: Change column name 'email' from table 'data_user' to 'personal_email'

```
35 •  ALTER TABLE data_user
36    RENAME COLUMN email TO personal_email
37    ;
```

100% 2:37

Action Output

	Time	Action	Response
1	18:50:57	ALTER TABLE data_user RENAME COLUMN email TO per...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0

Step 5: Add row in the table 'data_user' with 'id' (9999) and the rest of the fields in blank. Necessary step to be able to create the FK constraint

```
43 •  INSERT INTO data_user (id)
44    VALUES (9999)
45    ON DUPLICATE KEY UPDATE id = id
46    ;
```

100% 2:31

Action Output

	Time	Action	Response	Durat
2	19:38:47	INSERT INTO data_user (id) VALUES (9999) ON DU...	1 row(s) affected	0.042

Step 6: Create de FOREIGN KEY constraint between table ‘transaction’ and ‘data_user’

```sql
ALTER TABLE `transaction`
ADD CONSTRAINT FK_transaction_datauser
FOREIGN KEY (user_id) REFERENCES data_user(id)
;
```

100% 44:49

Action Output

	Time	Action	Response
1	19:44:03	ALTER TABLE transaction ADD CONSTRAINT FK_tra...	100000 row(s) affected Records: 100000 Duplic

Step 7: Delete the 'website' column from the 'company' table

```sql
ALTER TABLE company
DROP COLUMN website
;
```

100% 2:60

Action Output

	Time	Action	Response
1	19:54:28	ALTER TABLE company DROP COLUMN website	0 row(s) affected Records: 0 Duplicates: 0 Warning

Step 8: Change the length of the 'credit_card_id' column of the 'transaction' table

```sql
ALTER TABLE `transaction`
MODIFY COLUMN credit_card_id VARCHAR(20)
;
```

100% 2:67

Action Output

	Time	Action	Response
1	20:28:24	ALTER TABLE `transaction` MODIFY COLUMN credi...	0 row(s) affected Records: 0 Duplicates: 0 W

Step 9: On the table 'credit_card' —> increase column length 'id'

```
ALTER TABLE credit_card
MODIFY COLUMN id VARCHAR(20)
;
```

100% 35:70

Action Output

	Time	Action	Response
1	20:36:30	ALTER TABLE credit_card MODIFY COLUMN id VAR...	0 row(s) affected Records: 0 Duplicates: 0 Warnin

Step 10: On the table 'credit_card' —> change ‘pin’ data type

```
ALTER TABLE credit_card
MODIFY COLUMN pin VARCHAR(4)
;
```

100% 1:75

Action Output

	Time	Action	Response
1	20:41:42	ALTER TABLE credit_card MODIFY COLUMN pin VA...	5001 row(s) affected Records: 5001 Duplicates: 0...

Step 11: On the table 'credit_card' —> change ‘cvv’ data type

```
ALTER TABLE credit_card
MODIFY COLUMN cvv INT
;
```

100% 24:79

Action Output

	Time	Action	Response
1	20:45:13	ALTER TABLE credit_card MODIFY COLUMN cvv INT	5001 row(s) affected Records: 5001 Duplicates: 0..

Step 12: On the table 'credit_card' —> change ‘expiring_date’ data type

```
ALTER TABLE credit_card
MODIFY COLUMN expiring_date VARCHAR(20)
;
```

100% 3:92

Action Output

	Time	Action	Response
1	20:49:39	ALTER TABLE credit_card MODIFY COLUMN expirin...	5001 row(s) affected Records: 5001 Duplicate

Step 13: On the table 'credit_card' —> create new column ‘fecha_actual’

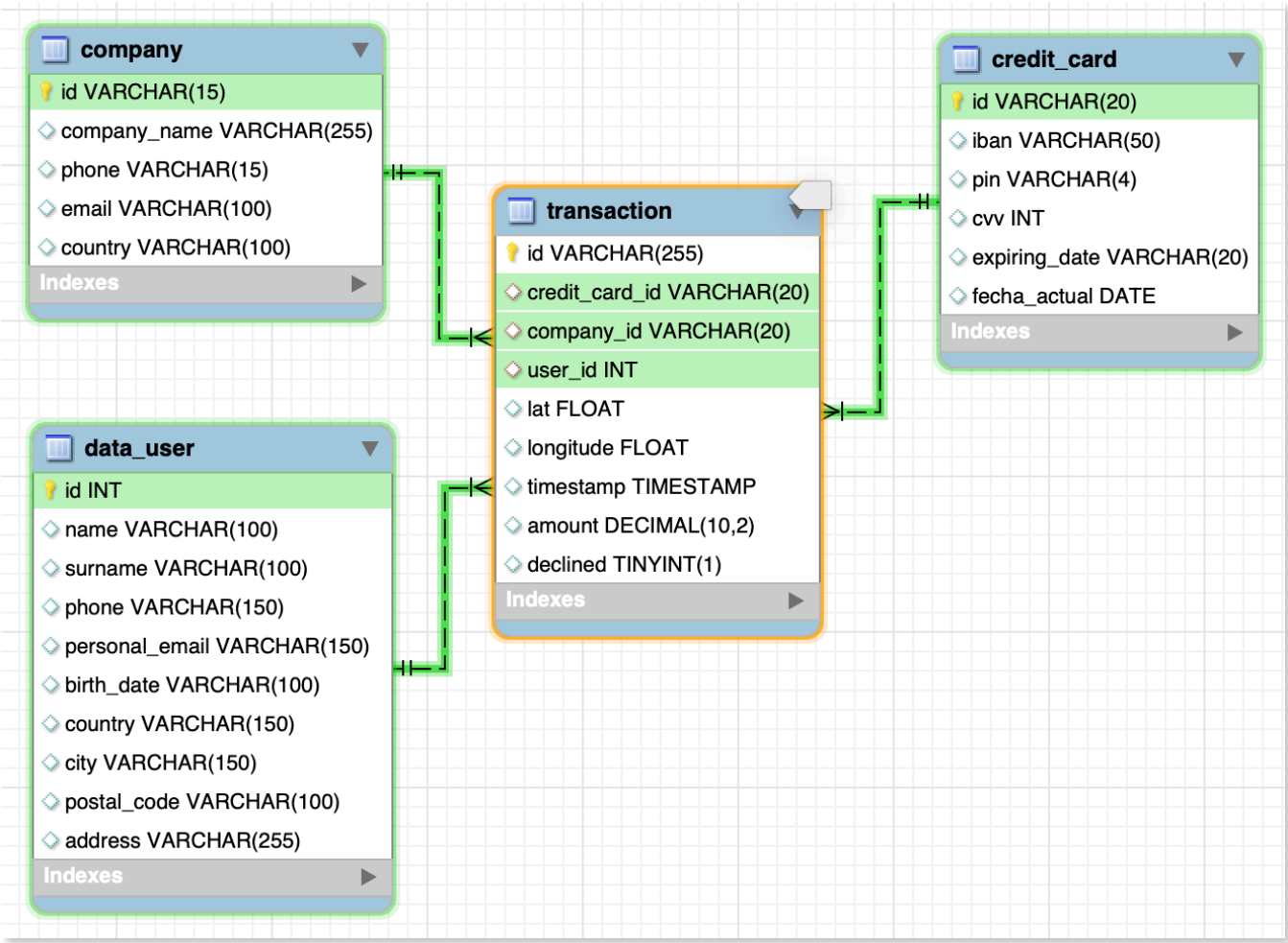
```
ALTER TABLE credit_card
ADD fecha_actual DATE
;
```

100% 1:96

Action Output

	Time	Action	Response
1	20:54:22	ALTER TABLE credit_card ADD fecha_actual DATE	0 row(s) affected Records: 0 Duplicates: 0 Warnir

Below you can see the final diagram result of applying all these SQL scripts on the transaction database.



As a final note of exercise 1, I want to explain that I have made the DDL (Data Definition Language - structure modification) and DML (Data Manipulation Language - data modifications) of this exercise step by step, because the statement specified it.

But it would be much more efficient, grouping the DDL by tables with an ALTER per table and the DMLs also separately.

In this case, we cannot use TCL (Transaction Control Language) START TRANSACTION - COMMIT; since DDL makes COMMIT.

Below there is an example of how the exercise would look focusing on efficiency:

```sql
-- USER
ALTER TABLE `user`
RENAME TO data_user,
MODIFY COLUMN id INT,
RENAME COLUMN email TO personal_email
;
INSERT INTO data_user (id)
VALUES (9999)
ON DUPLICATE KEY UPDATE id = id
;
-- TRANSACTION
ALTER TABLE `transaction`
MODIFY COLUMN credit_card_id VARCHAR(20),
ADD CONSTRAINT FK_transaction_datauser
FOREIGN KEY (user_id) REFERENCES data_user(id)
;
-- COMPANY
ALTER TABLE company
DROP COLUMN website
;
-- CREDIT_CARD
ALTER TABLE credit_card
MODIFY COLUMN id VARCHAR(20),
MODIFY COLUMN pin VARCHAR(4),
MODIFY COLUMN cvv INT,
MODIFY COLUMN expiring_date VARCHAR(20),
ADD COLUMN fecha_actual DATE
;
```

- Exercise 2

L'empresa també us demana crear una vista anomenada "InformeTecnico" que contingui la següent informació:

ID de la transacció
Nom de l'usuari/ària
Cognom de l'usuari/ària
IBAN de la targeta de crèdit usada.

Nom de la companyia de la transacció realitzada.

Assegureu-vos d'incloure informació rellevant de les taules que coneixereu i utilitzeu àlies per canviar de nom columnes segons calgui.

Mostra els resultats de la vista, ordena els resultats de forma descendent en funció de la variable ID de transacció.

Apart from the columns requested by the exercise, I have added several columns that seemed relevant to me or that could give us relevant information:

- amount: allows us to know how relevant a transaction is.

- birthday: it would allow to find out the age of the customers

- company_country and user_country: would allow us to establish if there is any relationship between the place of residence of a user and if they buy from companies of the same country.

- card_expiration_date and operation_declined: It would allow to check if there is any relationship between cancellations and the expiration dates of credit cards.

On the next page, you'll find the script for the creation of the view requested.

```sql
CREATE VIEW informetecnico AS
SELECT  t.id AS transactionID,
        cc.iban,
        DATE(t.timestamp) AS transaction_date,
        t.amount,
        du.name AS user_name,
        du.surname AS user_surname,
        STR_TO_DATE(du.birth_date, '%b %e, %Y') AS birthday,
        du.city AS user_city,
        du.country AS user_country,
        c.company_name,
        c.country AS company_country,
        cc.expiring_date AS card_expiration_date,
        t.declined AS operation_declined
FROM `transaction` t
JOIN data_user du ON t.user_id = du.id
JOIN company c ON t.company_id = c.id
JOIN credit_card cc ON t.credit_card_id = cc.id
ORDER BY t.id DESC
;
```

100% 47:325

Action Output

	Time	Action	Response
1	02:20:41	CREATE VIEW informetecnico AS SELECT t.id AS tra...	0 row(s) affected

```sql
SELECT * FROM transactions.informetecnico;
```

100% 43:1

Result Grid Filter Rows: Search Export: Fetch rows:

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FFF867C9-17B5-4B1F-AFD9-F8023...	XX278446342932680979729426	2015-04-12	448.63	Fqlngd	Lvhfqyxi	1967-05-04	Birmingham	United Kingdom	Cras Vehicula Aliquet Industries	Netherlands	2025-10-26	0
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informetecnico 2

Action Output

	Time	Action	Response	Duration / Fetch Time
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